

CMP3103M AMR

Human-Robot Interaction

part 2

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Attendance Code:

680470

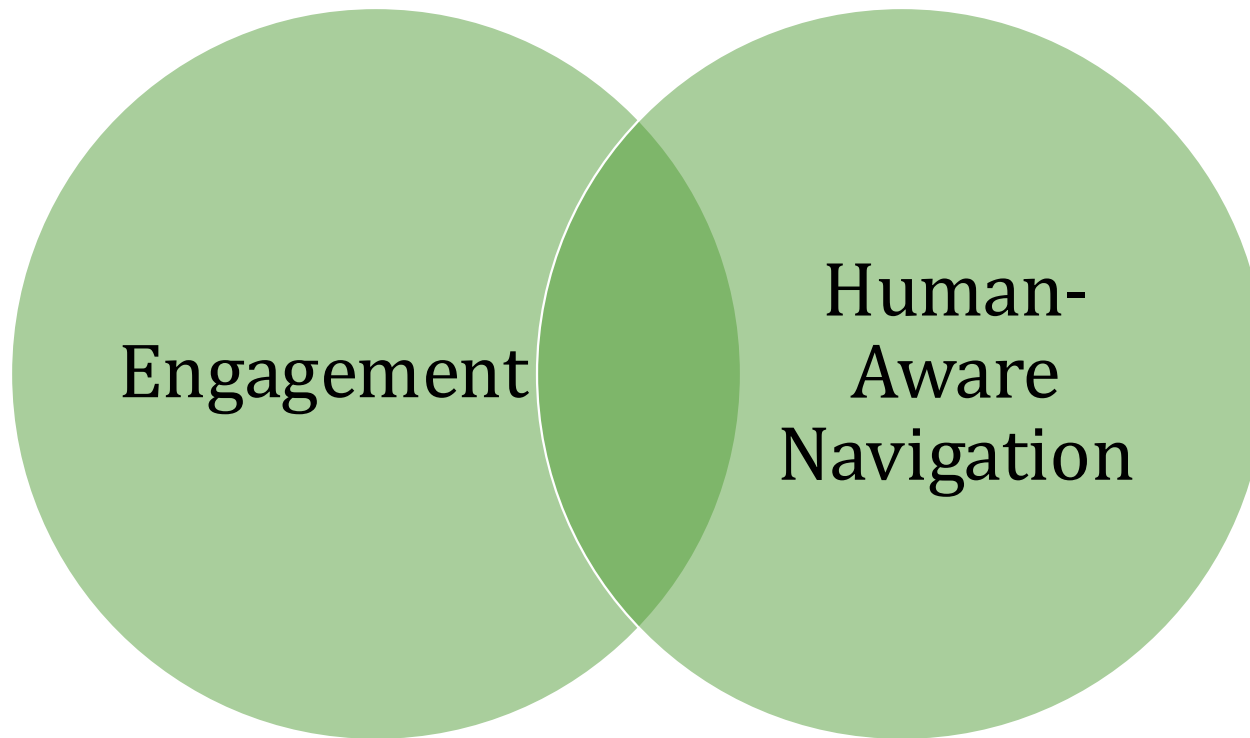
Guest Lecture TODAY

Harnessing Physical Interaction for Human-Robot Collaboration: How Robots Learn from Human Touch

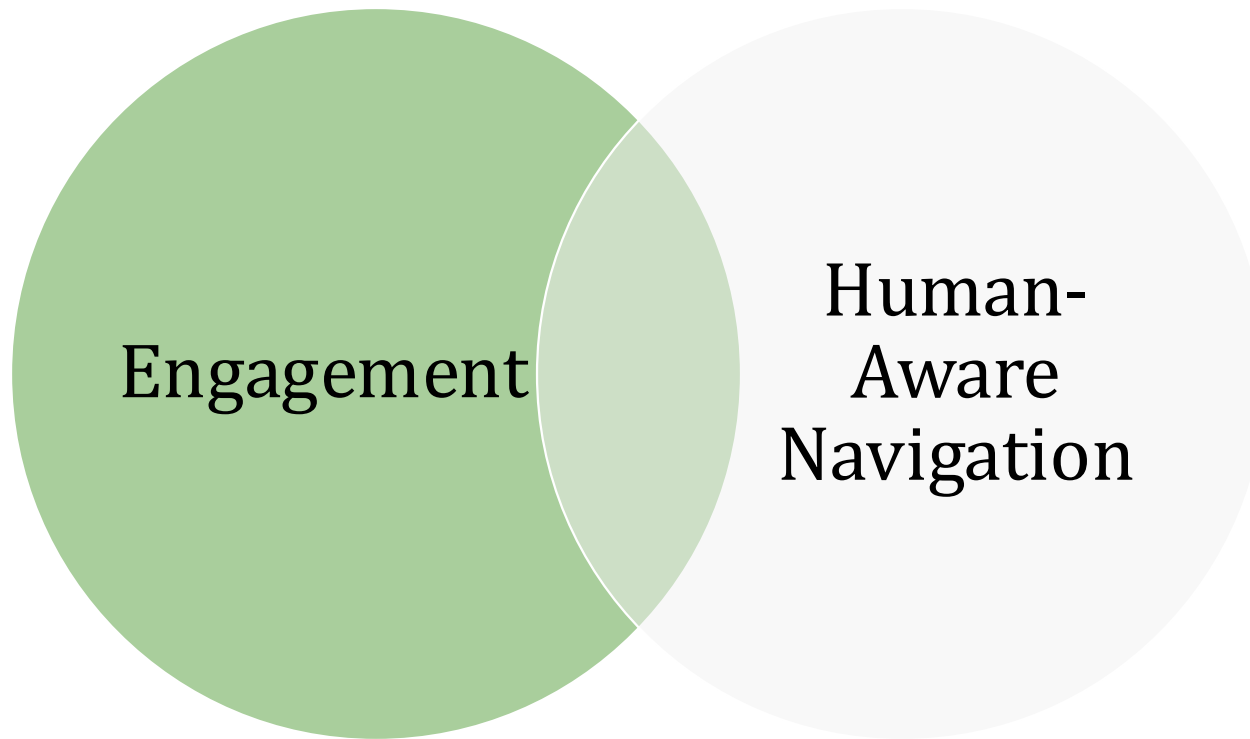
Dr Ayse Kucukyilmaz, University of Nottingham

Human-robot interaction is often framed in terms of vision, speech, and planning, but touch remains an underused yet profoundly expressive modality. In this talk, I will explore how physical interaction can serve as a dialogue between humans and robots, enabling more adaptive, proactive, and collaborative robotic behaviour. Drawing on our work in haptic shared control, I will talk about robots can interpret haptic data to act as proactive partners, dynamically adjusting their roles in collaboration. I will also discuss how physical conflicts and failures, rather than being obstacles, can serve as valuable learning opportunities, enhancing human-robot teamwork, particularly in scenarios requiring close physical interaction.

Today



Today



What is engagement?

- Many different definitions...

See e.g. Glas & Pelachaud, 2015 for an overview of these

- Sidner et al 2005:

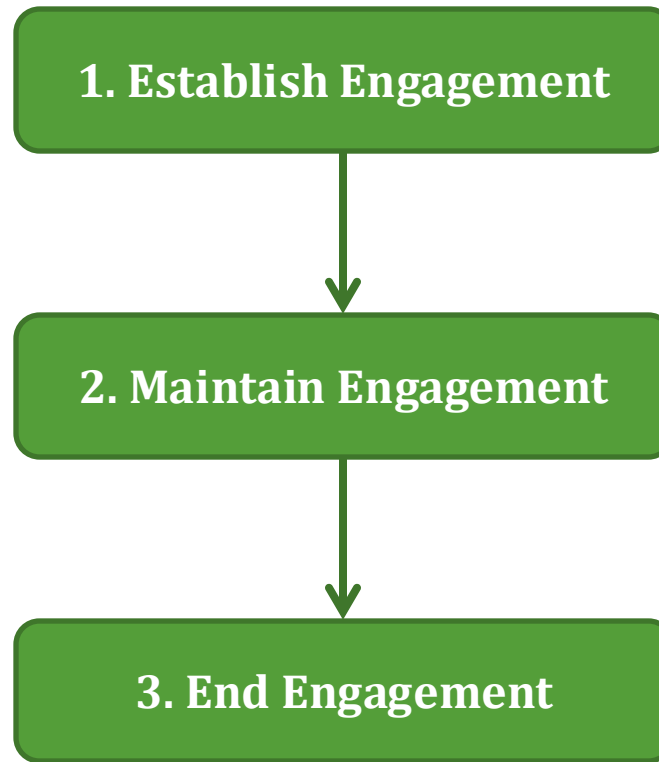
“...the process by which two (or more) participants establish, maintain and end their perceived connection during interactions they jointly undertake.”

- It encompasses all types of behaviour:

- Implicit/Explicit
- Verbal/Nonverbal



Stages in Engagement



1. Establish Engagement

- Attracting attention
- Drawing into an (ongoing?) interaction



The Collection
Art and Archaeology in Lincolnshire

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[Stop robot tasks](#) /room_task_server: 1 TASK_STARTED interfaceShow/home/start

The robot is recording encrypted data for scientific purposes. Ask for the leaflet at the reception desk of the museum to read the Data Protection Statement.

The Collection
Art and Archaeology in Lincolnshire

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Please, select one of the available tours.

Death and Burial tour	Tools and technology tour	Religion and belief tour	Art and design tour
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[Start this tour!](#)

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Bronze Age barrow finds

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The robot is recording encrypted data for scientific purposes. Ask for the leaflet at the reception desk of the museum to read the Data Protection Statement.



1. Establish Engagement

Social Strategies

- Speech
 - Saying something to someone
- Gesture
 - Only useful if have limbs...
 - Waving, etc
- Behaviour
 - Could engage in attention seeking behaviour
- Physical Interaction
 - Equivalent of a tap on the shoulder... (clearly safety implications)

Non-Social Strategies

- Sound
 - Pre-recorded speech
 - Alarm / beeps / motor noise
 - White noise
- Vision
 - Flashing lights
- Behaviour
 - Bumping into human (not advisable...)

Video from:

<https://www.youtube.com/watch?v=asHaWo04mIo>

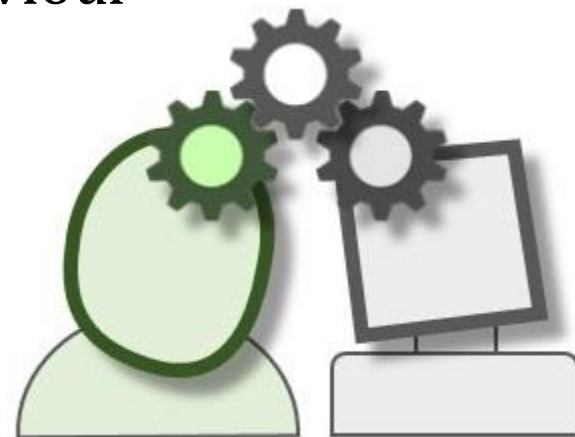


2. Maintain Engagement

- Recall that we are at least partially relying on:
 - Anthropomorphism: appearance and behaviour
 - Attribution of agency

This can only get you so far!

- The necessity for going beyond this with “deeper” complexity and adaptivity of behaviour
 - Refer to Control Architectures Lecture
 - This also underlies part of the motivation for incorporating Cognitive Architectures into HRI systems



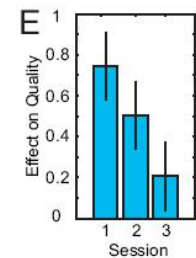
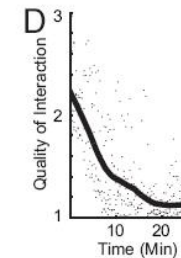
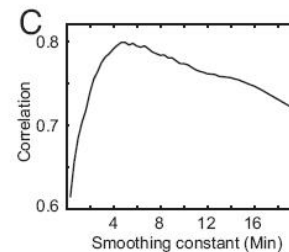
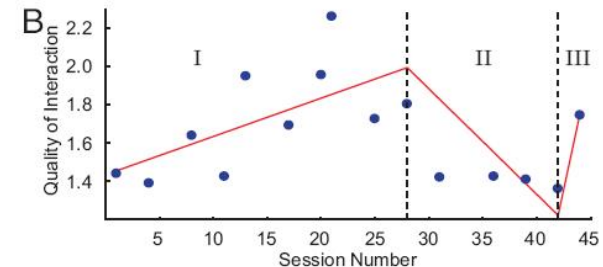
3. End Engagement

- Ending the interaction, and hence any engagement in the interaction
 - Possibly to be resumed at a later time
- Task related:
 - Joint task has completed successfully/unsuccessfully
 - Task no longer relevant
- Personal related
 - Illusion of agency has gone
 - Boredom!



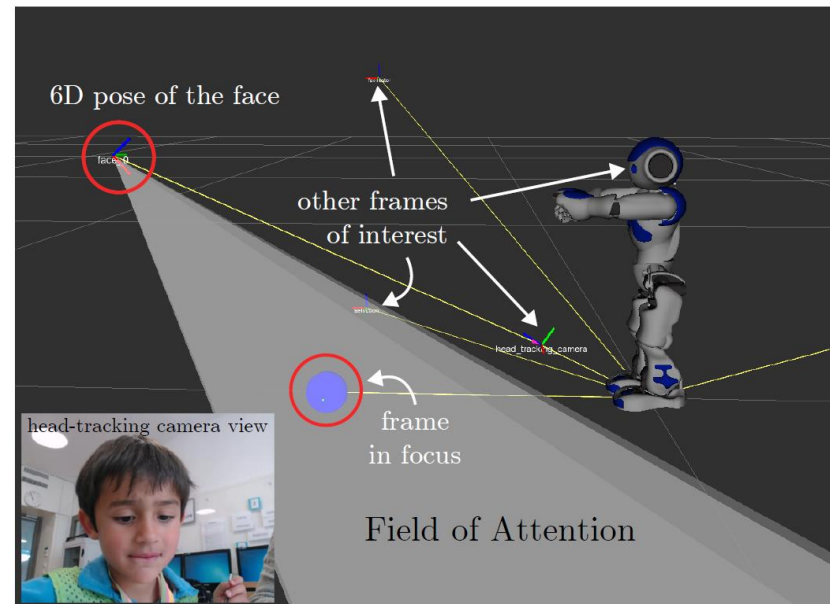
How to detect Engagement?

- How to tell whether someone is engaged in an interaction?
 - And, ideally, how to do so automatically?
 - What should you examine?
- This is a difficult task!
- Humans have an intuitive sense of engagement, based on extensive experience:
 - Developed from baby onwards
 - Can take advantage of this
 - E.g. Tanaka et al (2007): rating using a 'slider'



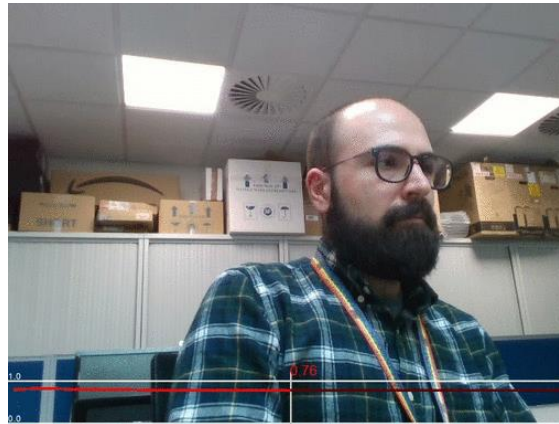
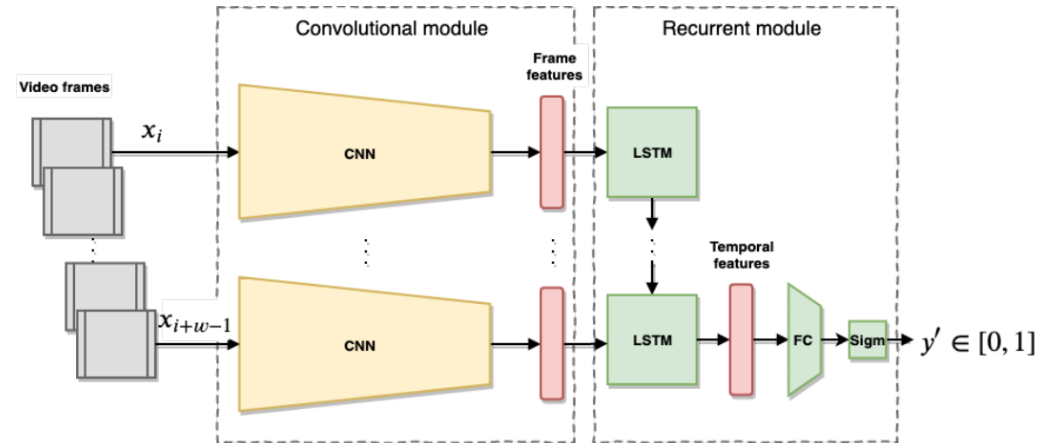
How to detect Engagement?

- Gaze is often used as an indicator of engagement
 - E.g. Baxter et al, 2014
- The problem is that this is often post hoc analysis
 - I.e. not in real-time, but only afterwards
- Recent developments trying to achieve this in real-time, using robot sensors:
 - E.g. the GAZR gaze estimator, which can be used for an estimation of engagement (Lemaignan et al, 2016)
 - ROS package:
<https://github.com/severin-lemaignan/gazr>



How to detect Engagement?

- Rather than using a model (gaze), use machine (deep) learning
 - E.g. del Duchetto et al, 2020
- Engagement is modelled as a continuous value between 0 and 1





Lindsey in the Museum 2018-2023

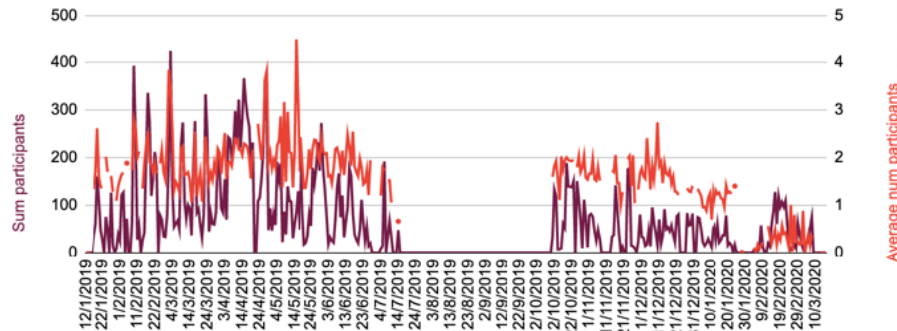
<https://lcas.lincoln.ac.uk/wp/projects/lindsey-a-robot-tour-guide/>

Reinforcement learning in the public domain?

- Lindsey is a tour guide robot in The Collection Archeological Museum
- *Can we learn better tours through long-term interaction with the public?*



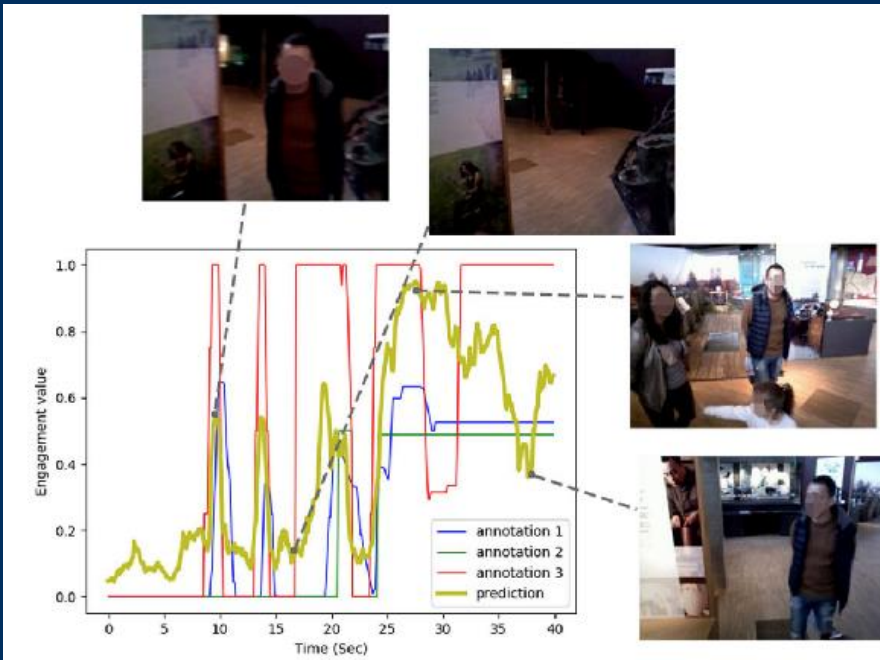
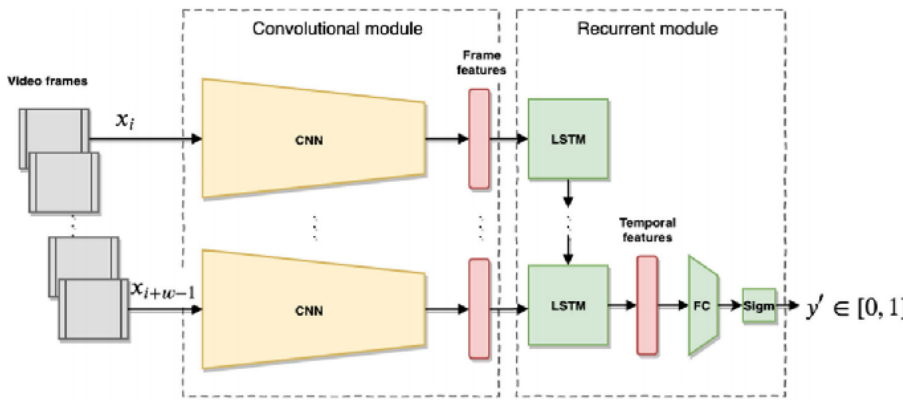
Tour participants per day



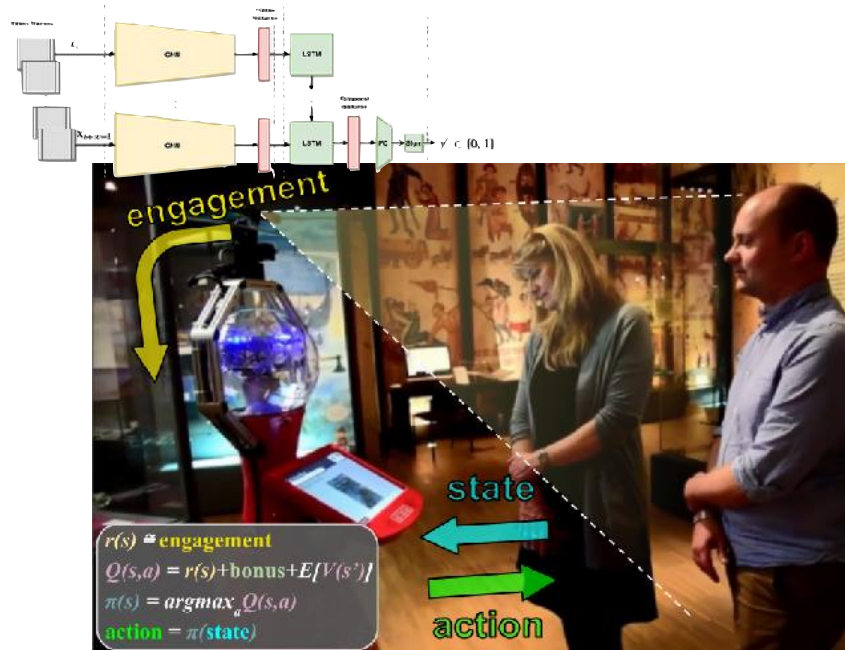
Are You Still With Me? Continuous Engagement Assessment From a Robot's Point of View

Francesco Del Duetto*, Paul Baxter and Marc Hanheide

Lincoln Centre for Autonomous Systems, School of Computer Science, University of Lincoln, Lincoln, United Kingdom



Reinforcement learning in the public domain?



Getting better on the job!

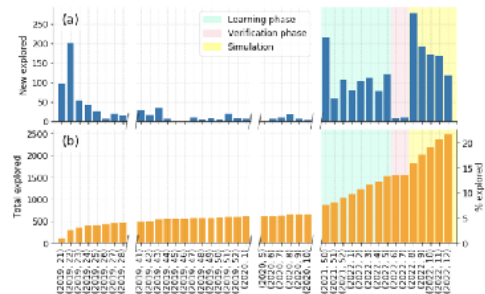
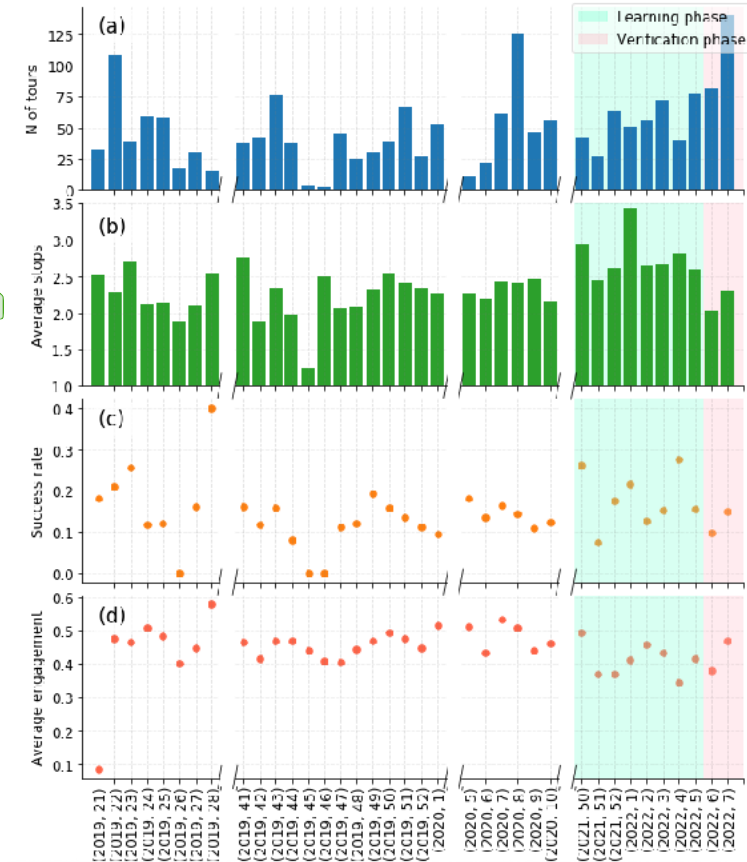
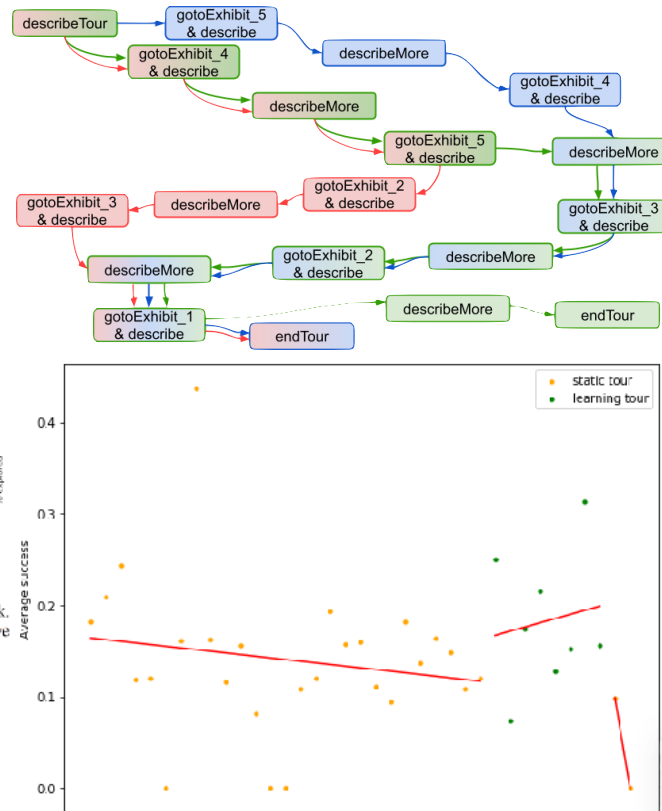


Fig. 9: Exploration of the state-action space of the policy per week. (a) Number of newly explored state-action pairs, (b) cumulative exploration.



Journals & Magazines > IEEE Robotics and Automation ... > Volume: 7 Issue: 3

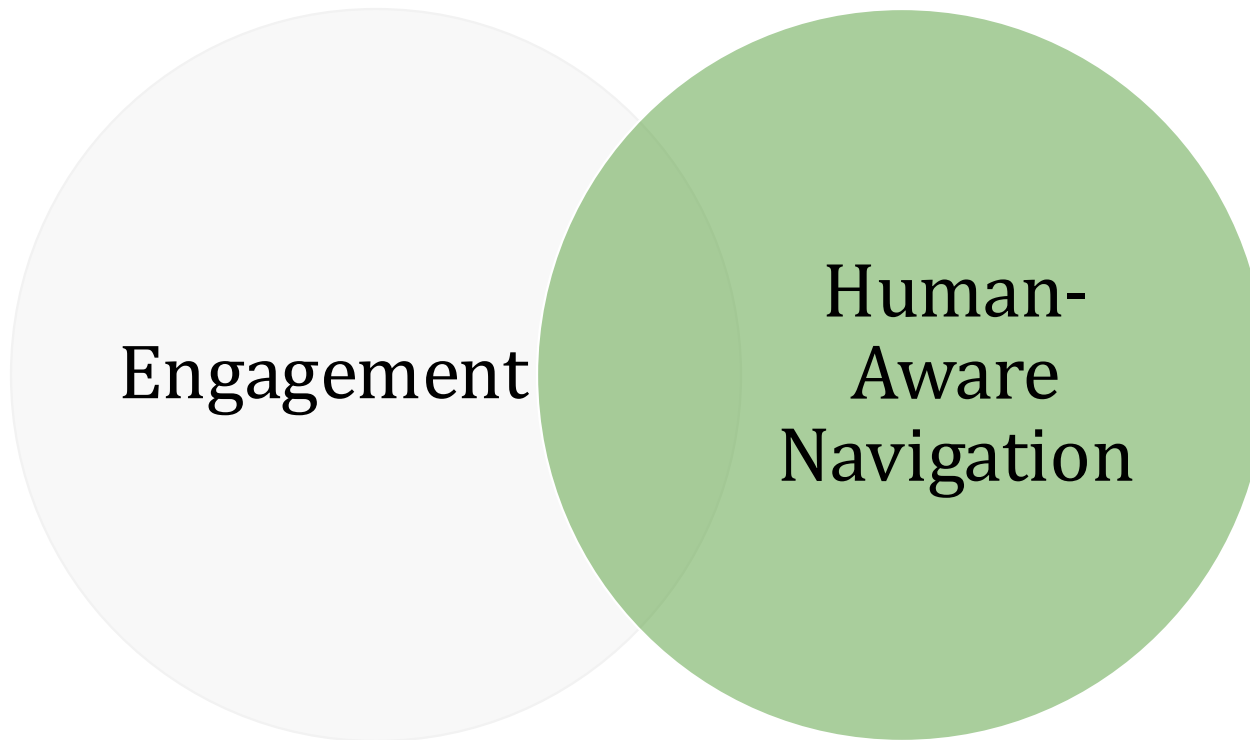
Learning on the Job: Long-Term Behavioural Adaptation in Human-Robot Interactions

Publisher: IEEE [Cite This](#) [PDF](#)

finalist of IROS 2022 Best Paper Award on Cognitive Robotics

Francesco Del Duchetto ; Marc Hanheide ; All Authors

Today

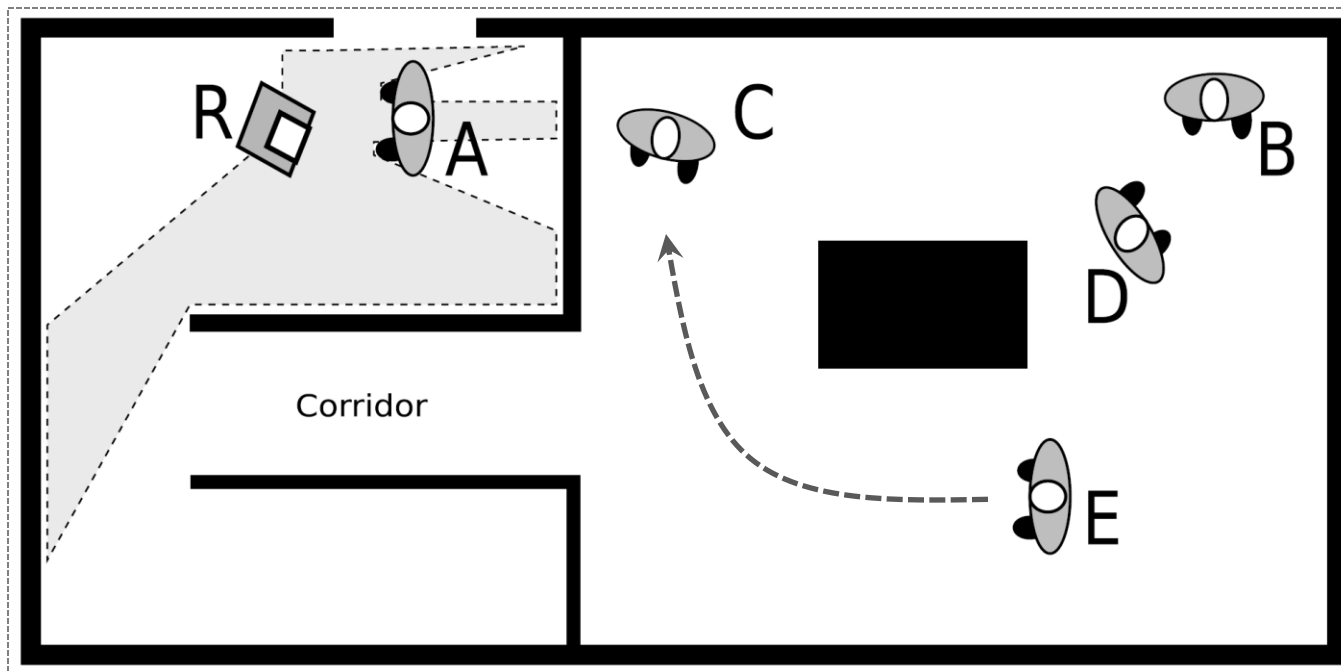


Human-Aware Navigation

- Autonomy for mobile robots requires navigation capabilities
- Obstacle avoidance clearly required
 - Preventing robot damage
 - Human safety!
- Goals (Kruse et al, 2013):
 1. Comfort: absence of annoyance and stress for humans
 2. Naturalness: similarity of robot behaviour to humans
 3. Sociability: adherence to high-level cultural constraints
- May be necessary to reduce efficiency (in terms of speed/distance to goal) in the service of these goals

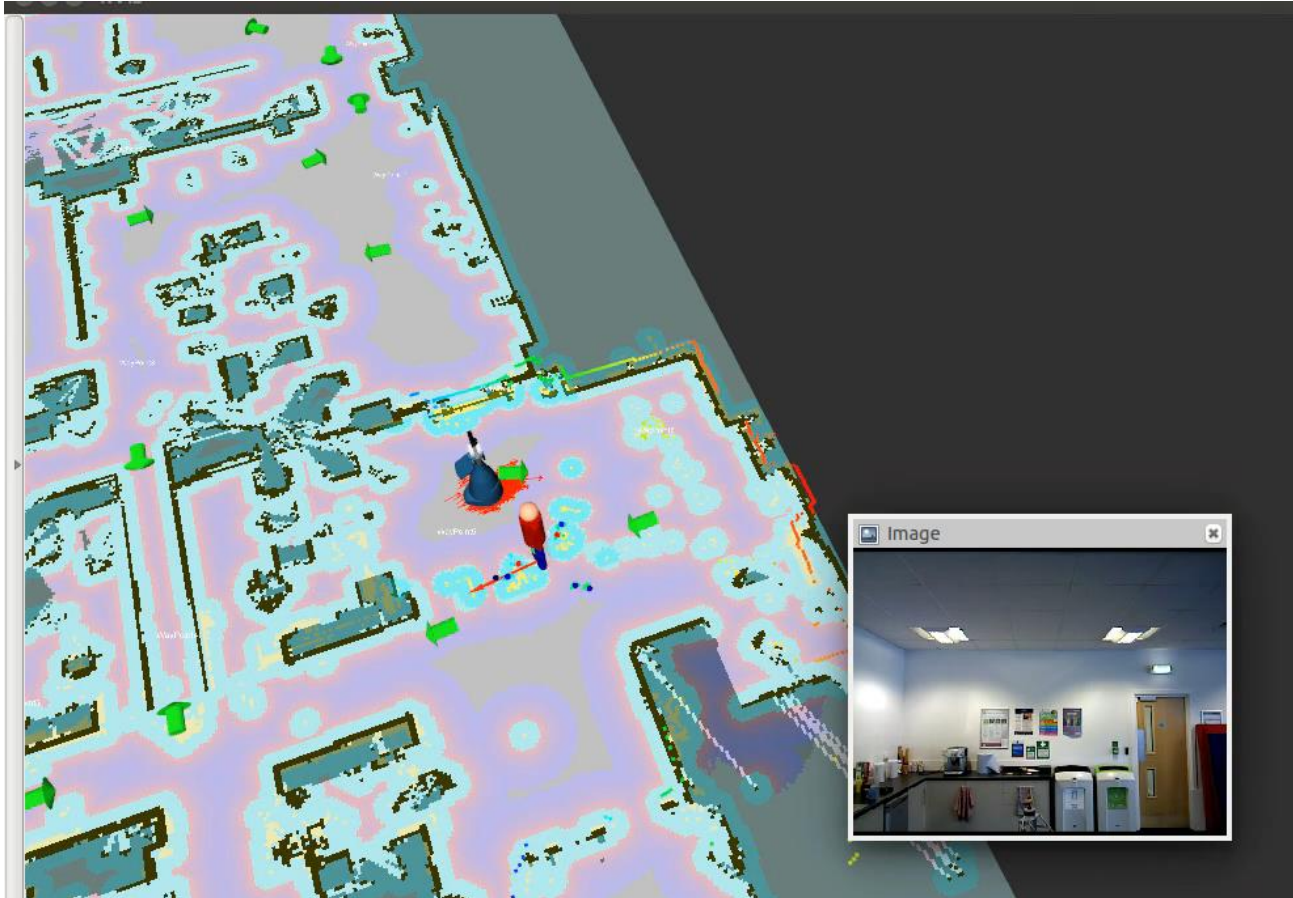
An example

From Kruse et al, 2013:



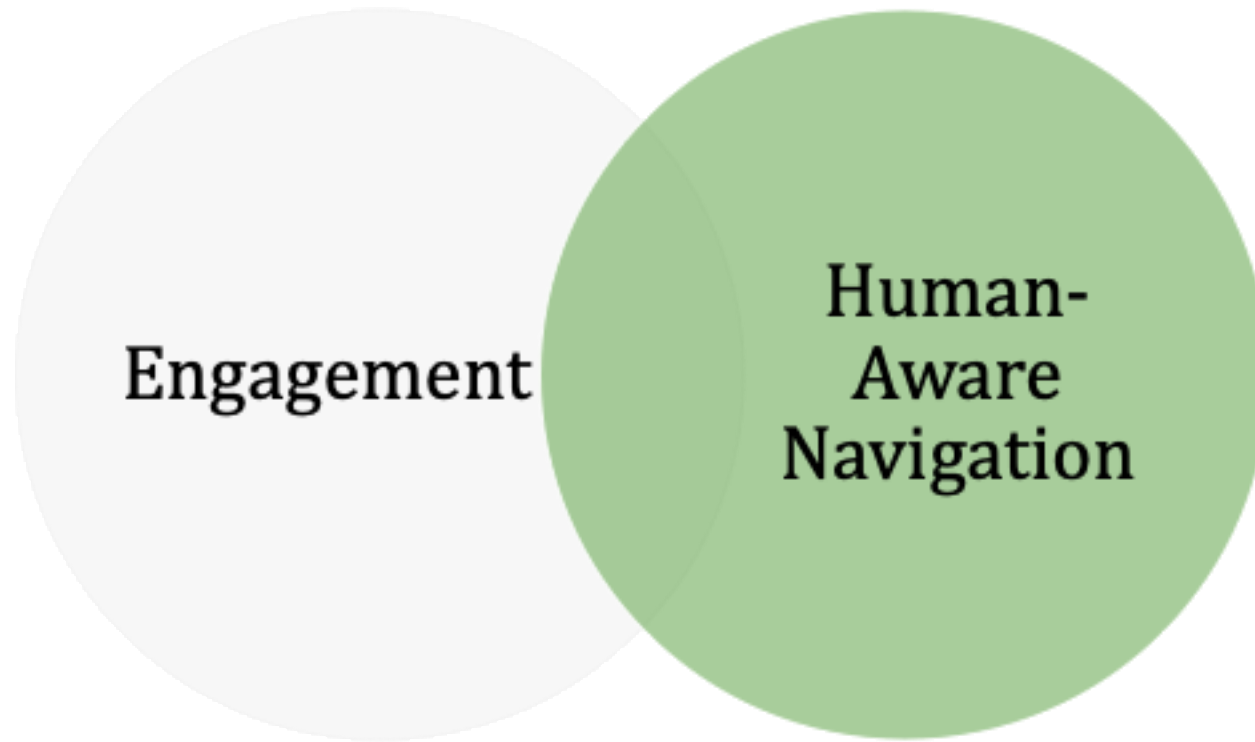
How should the robot guide person A to person B?

Humans are Awkward Obstacles...



Also see this L-CAS video:

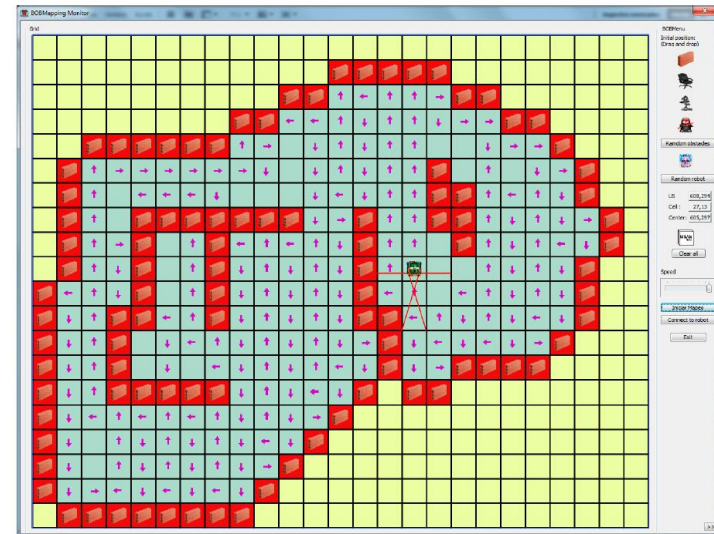
<https://www.youtube.com/watch?v=zdnvhQU1YNo>



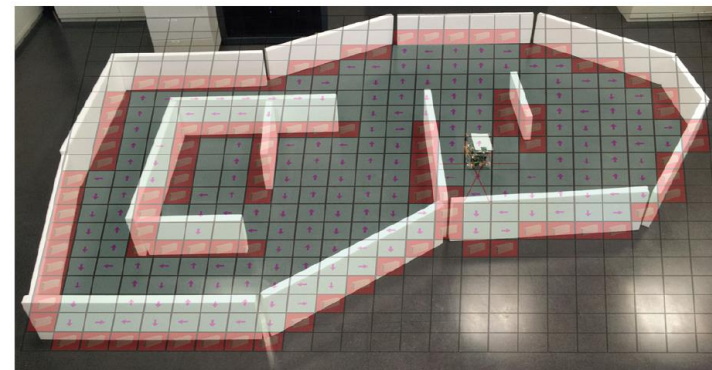
(a) Navigation and Costmaps

Recap: path planning

- Refresh your memory regarding Lecture Week on Path Planning
 - Dijkstra and A*
 - Include movement cost
- Application to discrete states, in this example a network of nodes
- Equally applies to a 2D space discretised in a grid
 - “Occupancy” grid
 - Image from (Gonzalez et al, 2013)



(a)



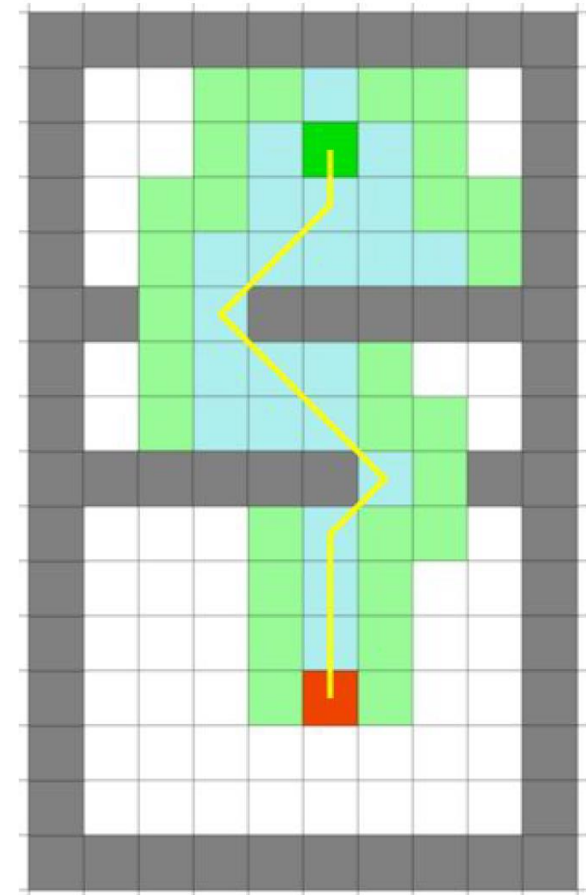
(b)

See <http://qiao.github.io/PathFinding.js/visual/>



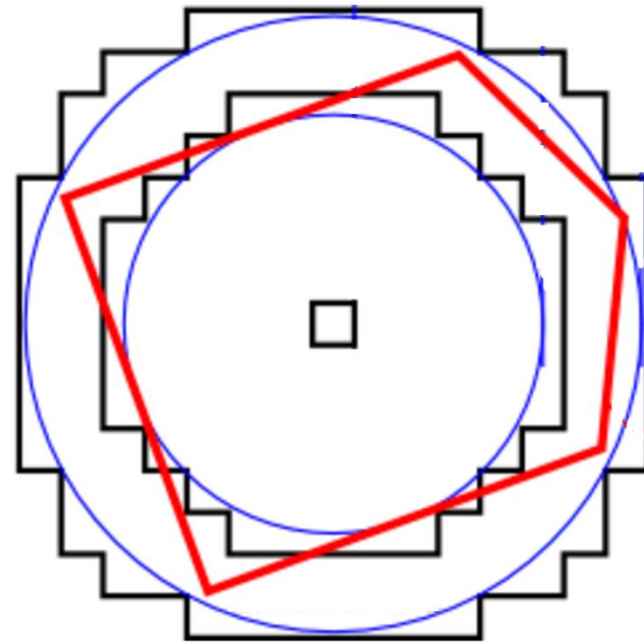
Path Planning

- Have what we need to plan path
 - E.g. Dijkstra
 - <https://qiao.github.io/PathFinding.js/visual/>
- However, Dijkstra not ideal:
 - Path close to obstacles
 - Next to walls
 - Not good for robots!
- Want to keep our robot safe:
 - If in corridor, drive in middle
 - Keep distance from obstacles



Robot Distances and Costs

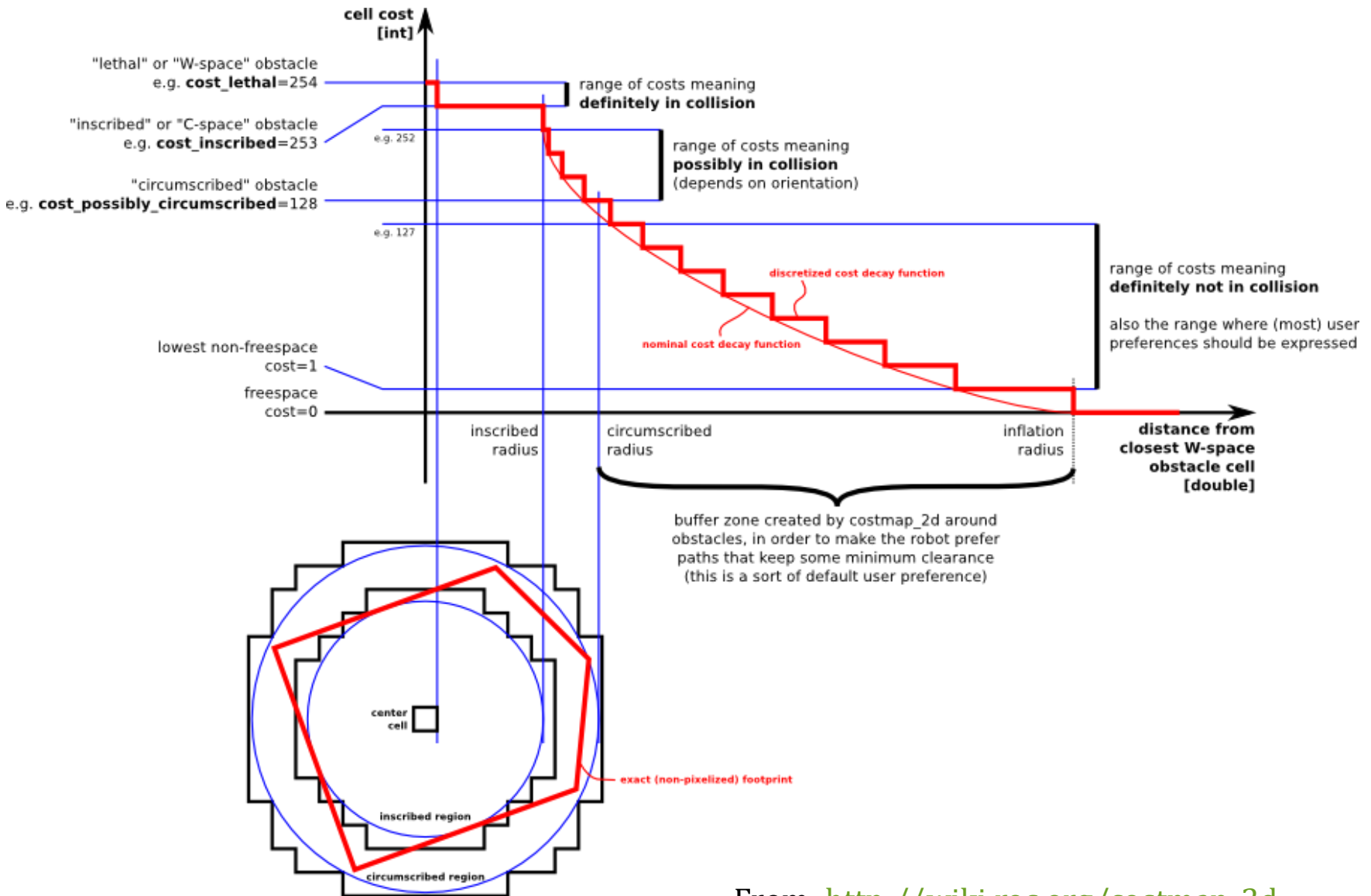
- Robot centre point assumed as point of rotation
 - E.g. the Limos
- Obstacles are “lethal”
 - In map
 - Sensed by laser
- Lethal obstacles “inflated” based on the robot size
 - Helps determine distance from obstacles



RED – actual robot footprint

OUTER CIRCLE –circumscribed region

INNER CIRCLE – inscribed region

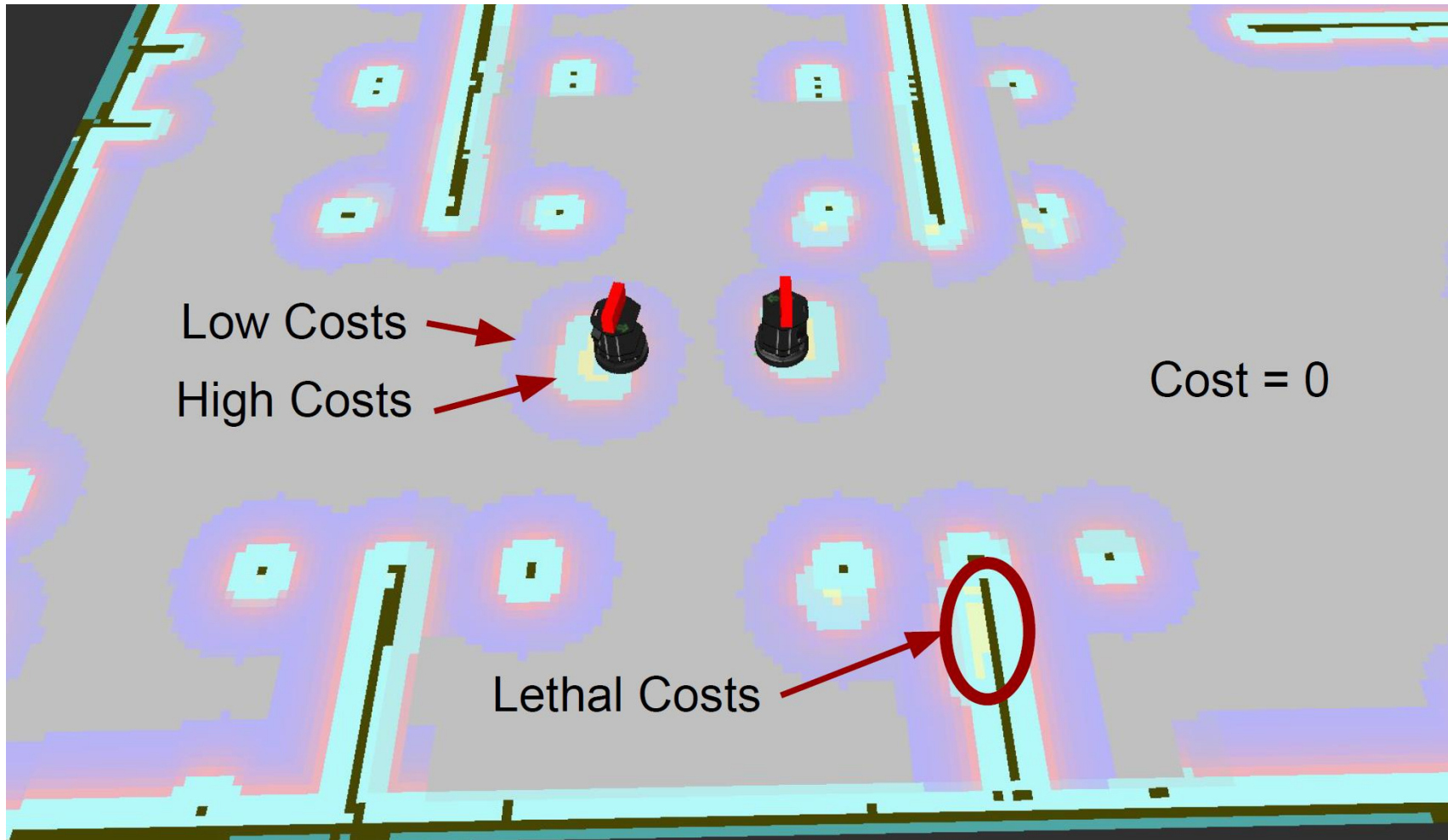


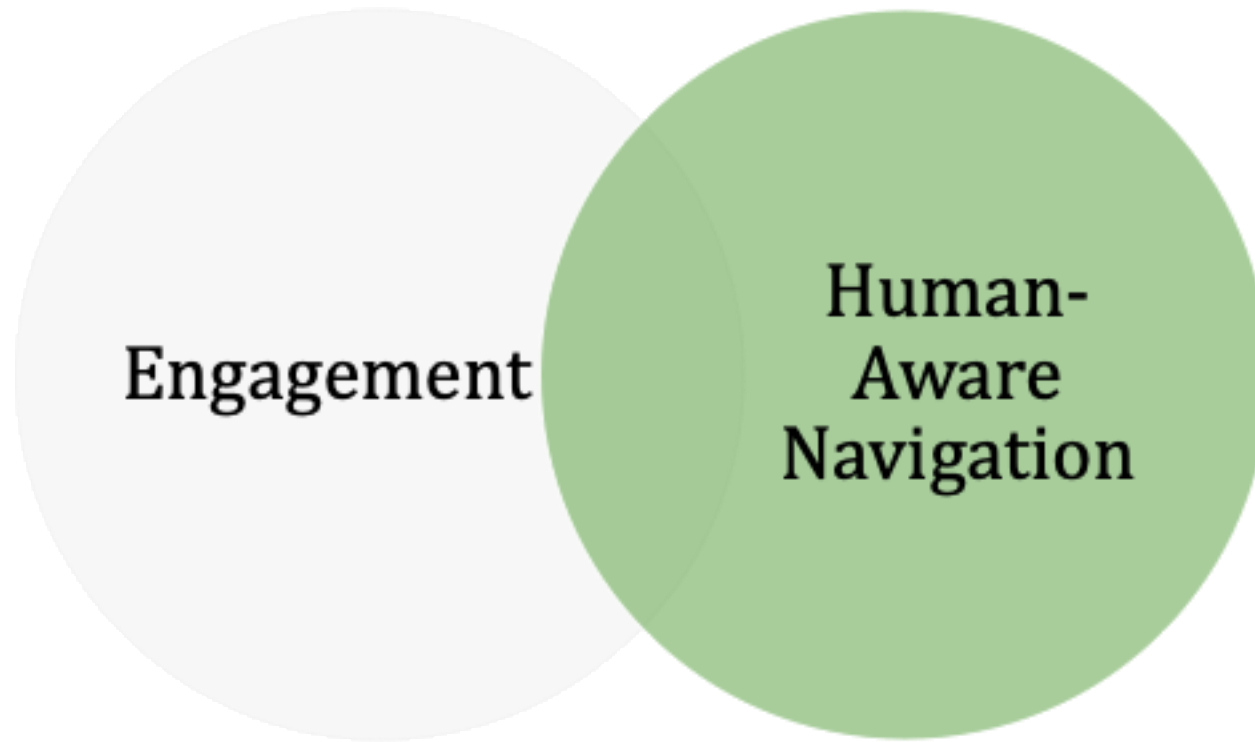
From: http://wiki.ros.org/costmap_2d

Using these costs

- Costs encoded into the cost-map, for each obstacle
- In Dijkstra example:
 - Using movement as part of the cost (1 for straight, $\sqrt{2} = 1.414$ for diagonal)
 - Now also use the cost of the relevant cells in the costmap to calculate the next neighbour
- Result:
 - For navigation, the robot can stay clear of obstacles, even when using Dijkstra for planning

Costmaps





(b) The Human Obstacle

Humans are not just obstacles...

- Human-Robot Spatial Interaction

The study of joint movement of robots and humans through space and the social signals governing these interactions

- Movement of robots and humans
- Focus on social signals

- Human-Aware Navigation

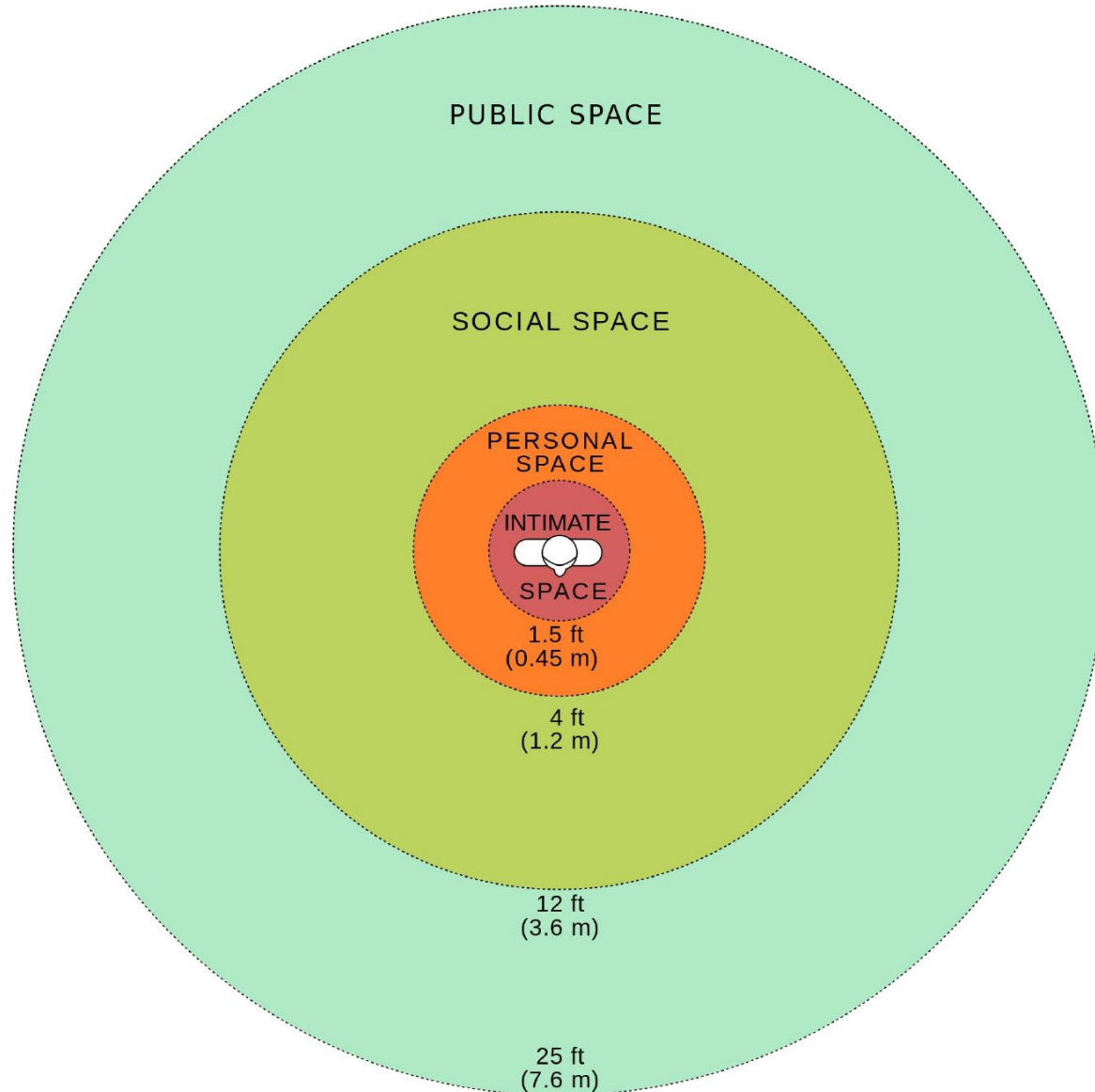
- Specifically taking the human into account
- Recall the three goals: comfort, naturalness, and sociability
- Two main methods:
 1. Stop-and-wait: human does all the hard work
 2. Cost functions based on principles of Proxemics

Proxemics

- A virtual personal space around an individual
 - Edward Hall, 1966
- Divided into four main zones
 - Each zone at a different distance, and with different interaction characteristics
 - Also dependent on relationship
 - Two 'phases' per zone

The four zones:

1. Intimate
2. Personal
3. Social
4. Public



Intimate Space

- 0 – 45cm: Intimacy
- Close Phase (0-15cm)
 - Intimacy, comforting
 - Vision blurred, vocalisations whispered
 - Senses of smell and radiant heat effective
 - Arms can encircle
- Far Phase (15-45cm)
 - Hand can reach and grasp extremities
 - Heads, thighs, and pelvis are not easily brought into contact
 - Able to focus the eye easily, peripheral vision includes the outline of the head and shoulders
 - Heat and odour of the other person's breath might be detected

Personal Space

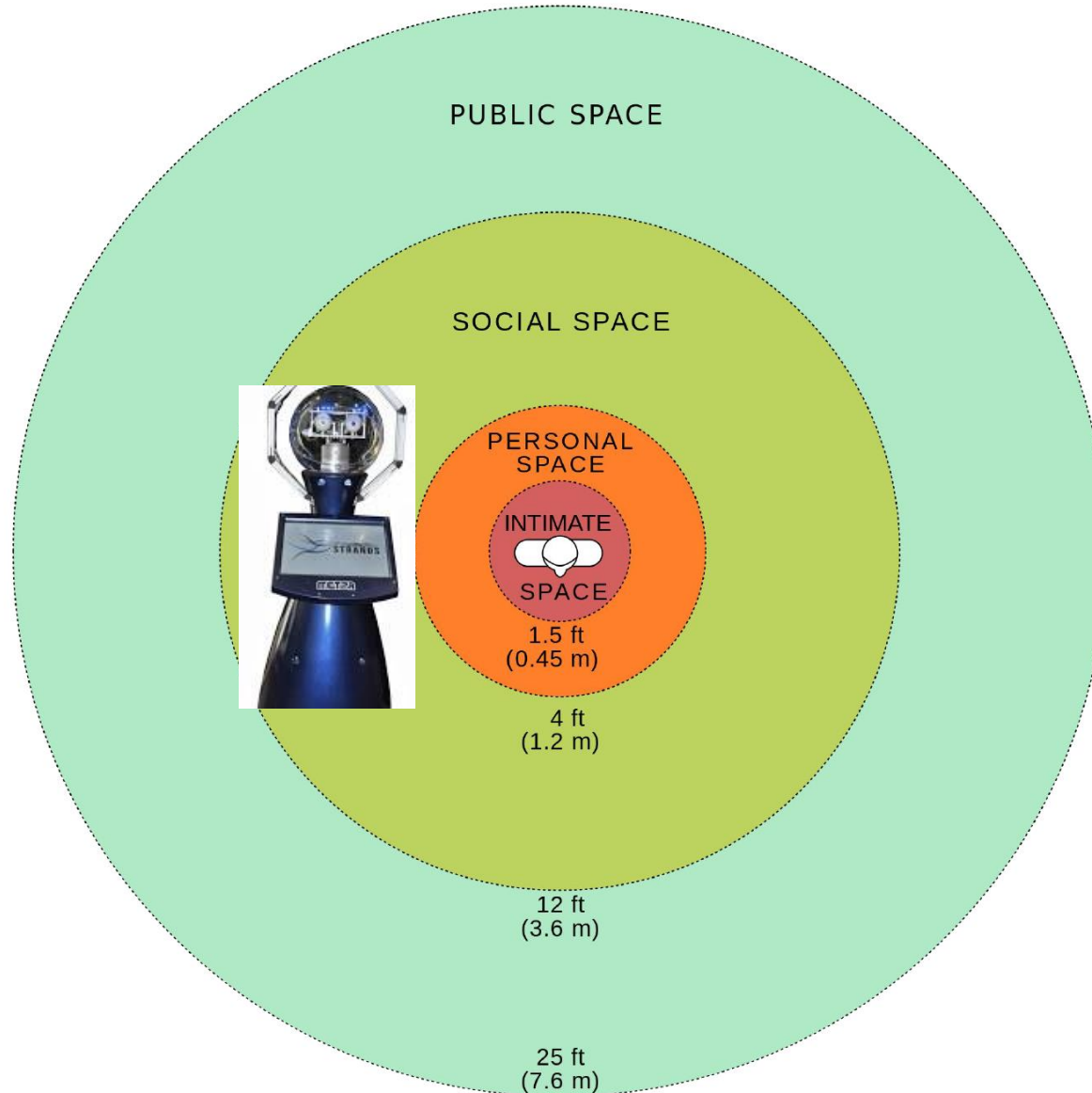
- 45 – 120cm: Family and good friends
- Close Phase (45-75cm)
 - Can grasp other person: peripersonal space
 - No visual distortion of other's features
 - Perception of 3-dimensional qualities of objects
- Far Phase (75-120cm)
 - Outside of grasping distance: at arm's length
 - Other's features clearly visible
 - Moderate voice volume
 - No perception of body heat
 - Lower levels of olfaction

Social Space

- 1.2m – 3.6m: Interactions with acquaintances/strangers
- Close Phase (1.2-2.1m)
 - No touching without special effort
 - Normal voice volume – can be heard from a moderate distance
 - Visual focus extends to nose and parts of both eyes, or, nose, mouth and one eye
- Far Phase (2.1-3.6m)
 - Fine details of face are lost
 - Skin texture, hair, teeth, and condition of clothes readily visible
 - Odour not detectable

Public Space

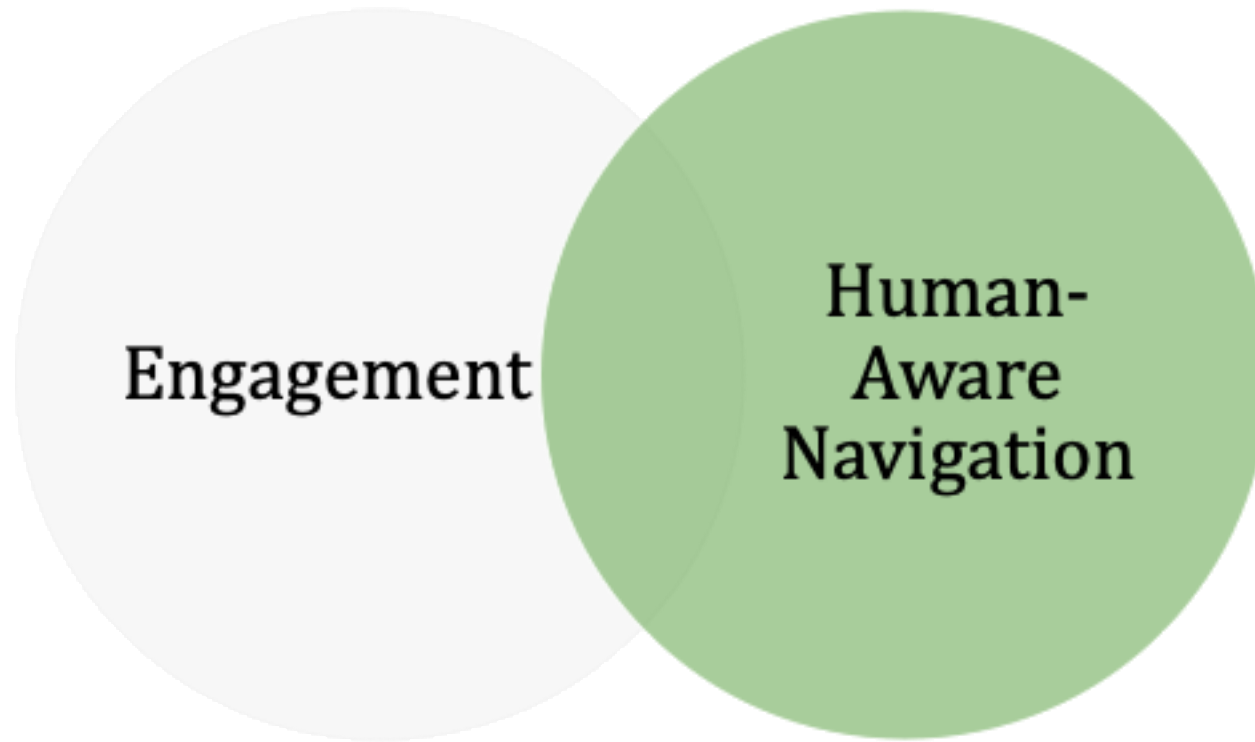
- > 3.6m: public speaking
- Close Phase (3.7-7.6m)
 - Can take evasive actions
 - Voice loud but not full volume
 - Can see whole face, fine details not visible
 - Only whites of eyes visible
- Far Phase (>7.6m)
 - Subtleties of meaning in voice is lost, as are details of facial expressions
 - Vocal, facial, and bodily expression must be exaggerated
 - Foveal vision takes in increasingly more of the other person



Caveats

- Dependent on culture (Hall's studies were in US)
- Equal spacing around individual
 - No difference between front and back
 - No accounting for movement (e.g. warping of space when walking)
- Do not take into account environmental conditions
 - E.g. dim lighting, loud environments, etc
- Enforced violations?
 - E.g. public transport
 - Changes behaviour
- These zones may be applicable to humans, but do they apply to robots?



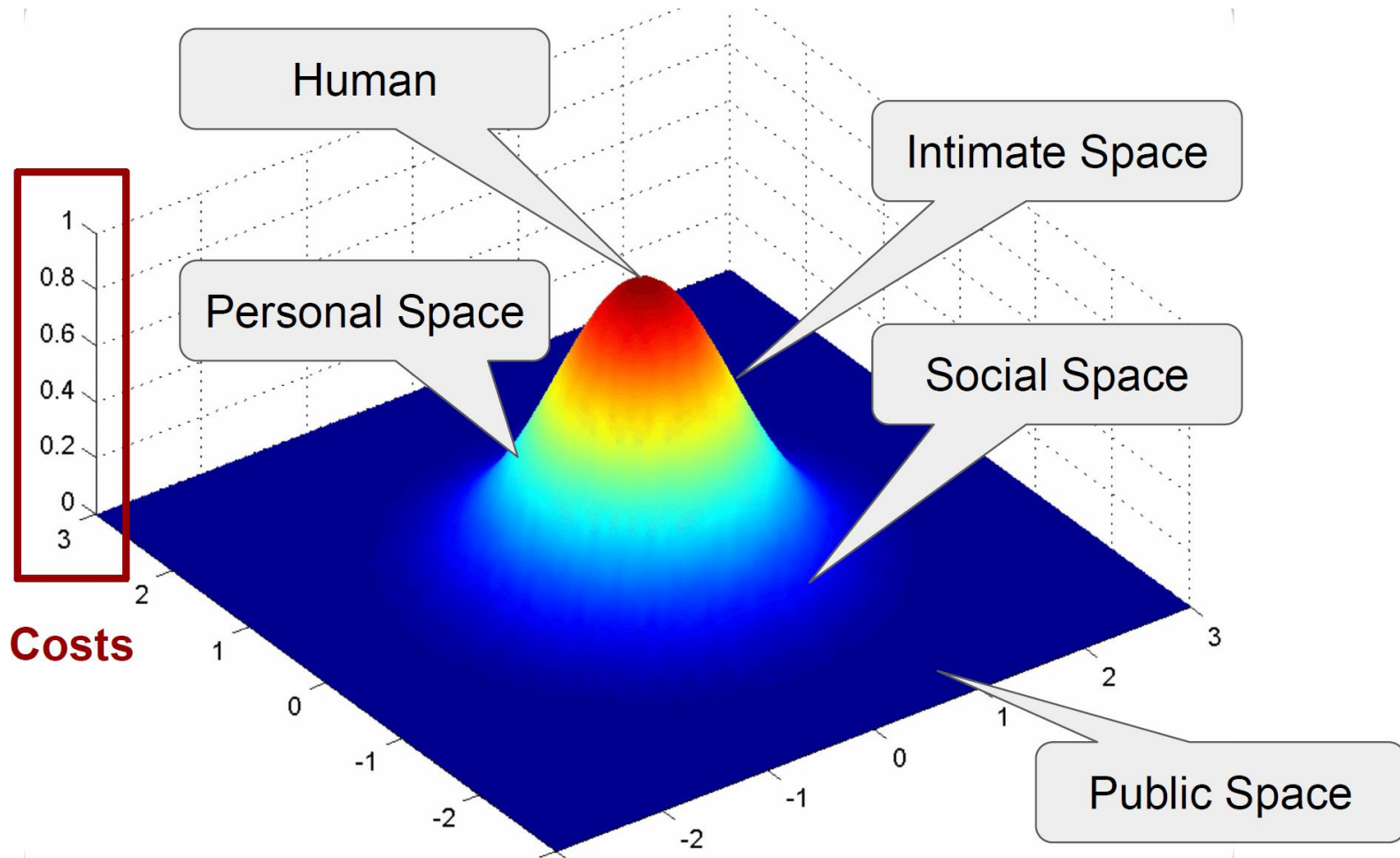


(c) Bringing These Together

Combining Costmaps and Proxemics

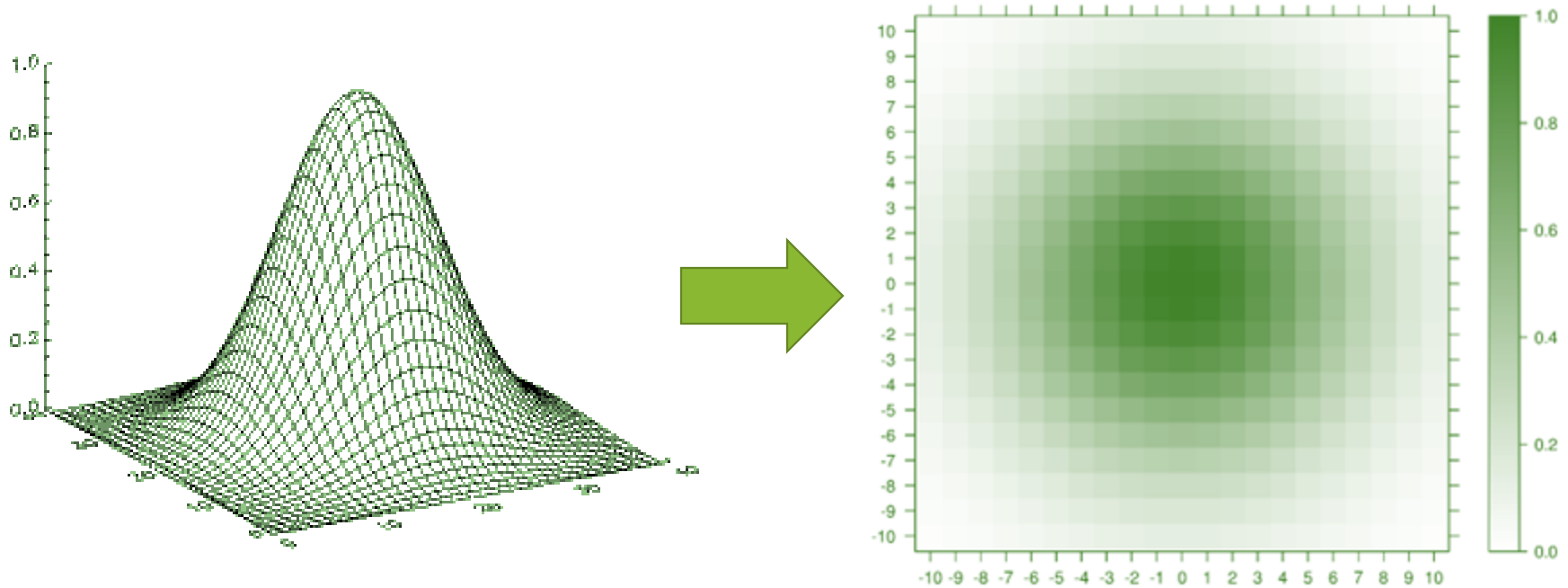
- Including costmaps as part of the human representation in the map:
 - Including proxemics as part of this
 - What is the benefit?
- Keeping a greater distance to the human
 - Perceived as safer
 - Reduced stress
 - Even though less “efficient”
- Not necessarily either of the other two goals
 - Doesn’t guarantee more natural behaviour
 - Doesn’t incorporate societal/cultural norms (e.g. drive on the left or the right?)
- How to put Proxemics into Costmap?

Proxemic Gaussian



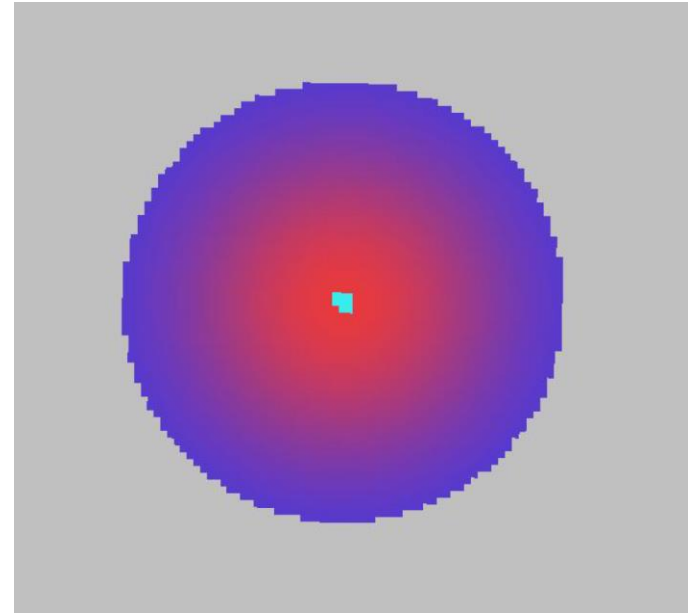
Discretisation

- Have a continuous function, need it to be discretised

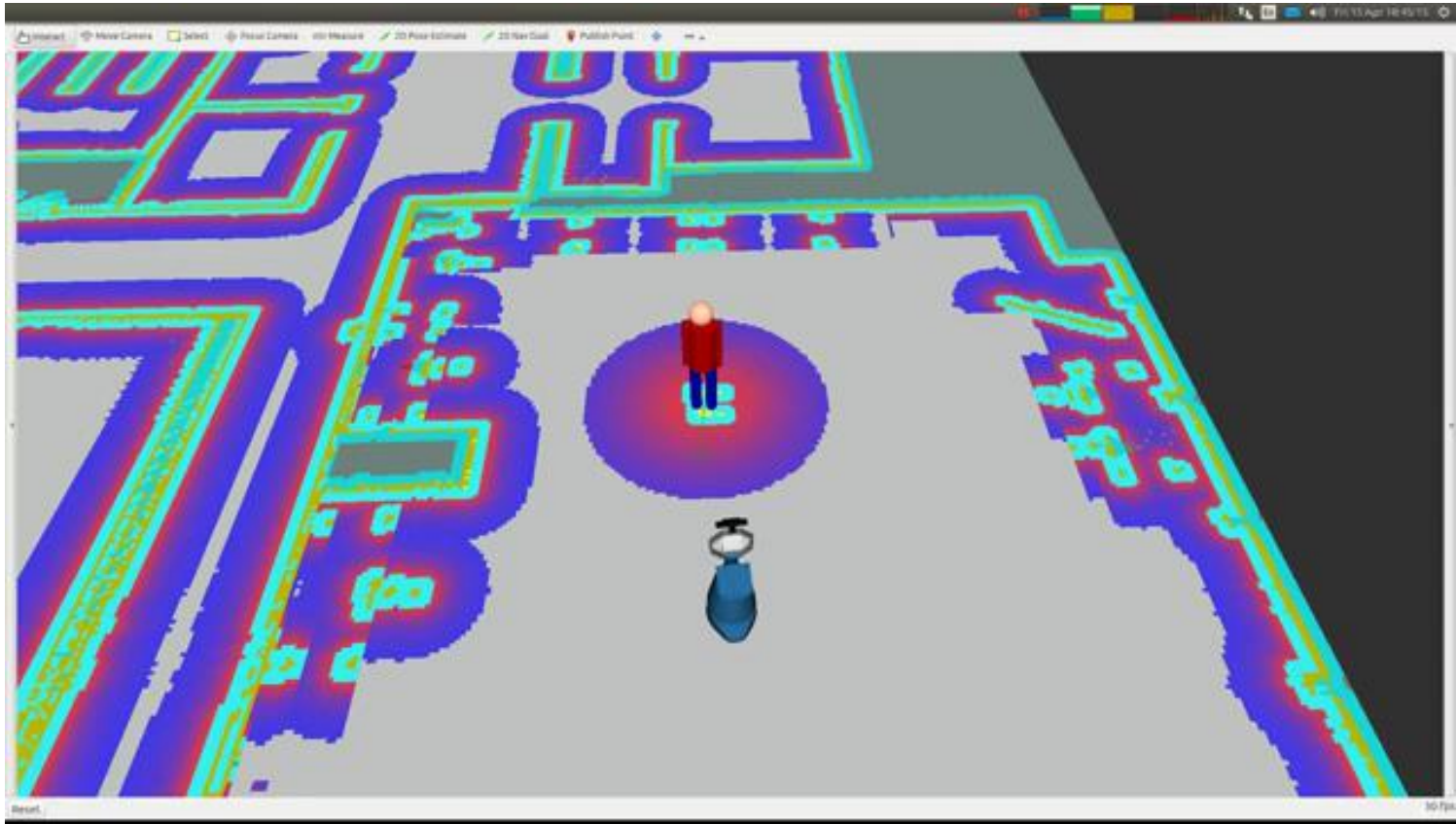


And back to Costmaps...

- This information can now be added to the overall costmap, with the other obstacles
 - Or added to a separate layer of the costmap (Lu et al, 2014)
- Path planning in this space as described before
 - Dijkstra
 - A*

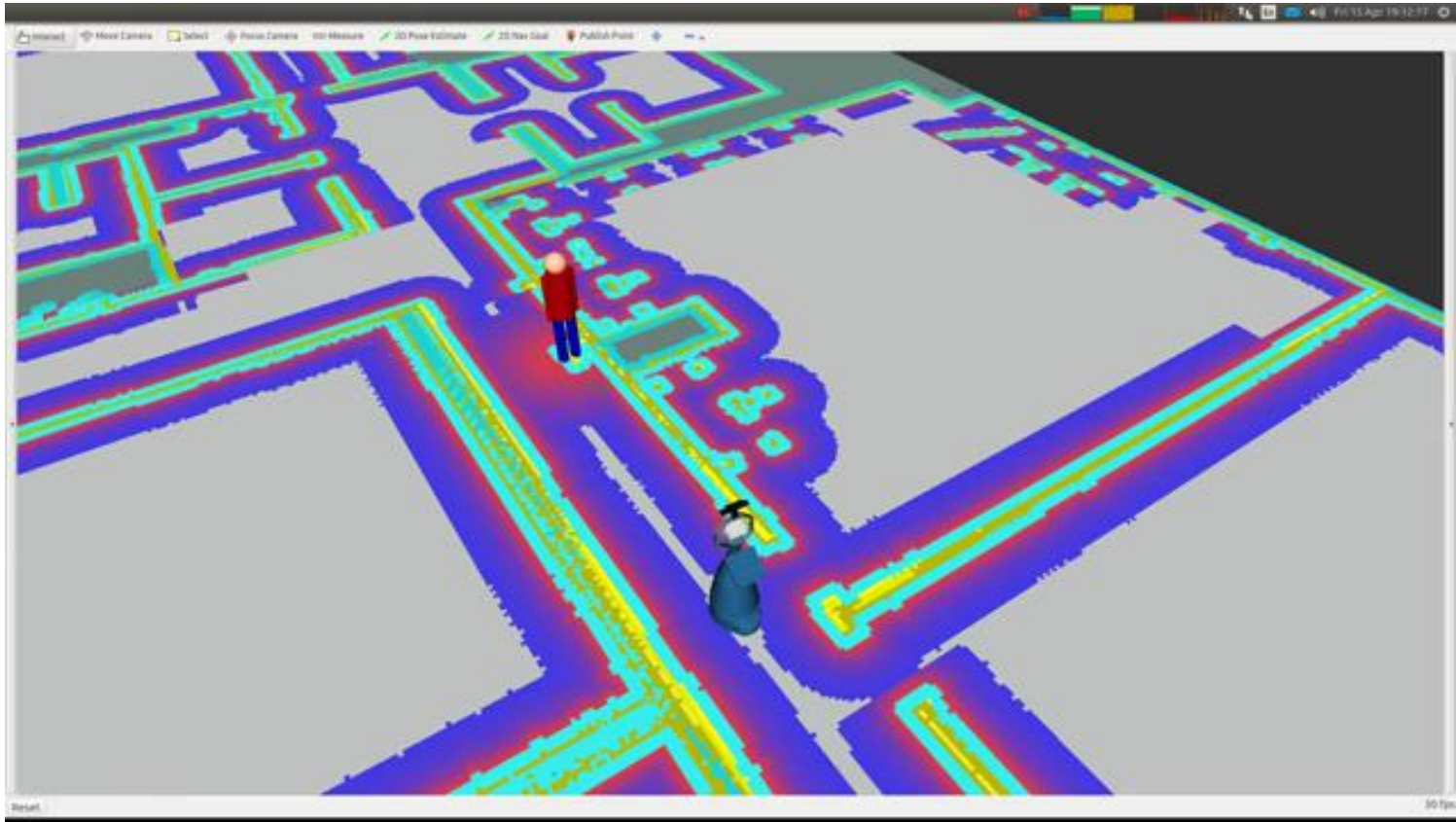


Navigation in Open Space



Full video: <https://www.youtube.com/watch?v=pg7g7qv80MU>

Navigation in Corridor



Full video: <https://www.youtube.com/watch?v=pg7g7qv80MU>

Gaussian Cost functions

Positives

- Straightforward implementation
- Is relevant for all environments
- Takes into account proxemics
- Ensures interaction is safe, and perceived to be as such

Negatives

- Only influences distances
- Sociability and naturalness not guaranteed
- Does not take into account social context
 - Only based on proxemics
 - Could drive through groups?

References / Reading

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